Identifiability of independent sampling units in publicly deposited metabolomics data: a repository-level audit and documentary assessment

Anas Enoch[1*]

[1*]Mohammed VI University of Sciences and Health (UM6SS), Casablanca, Morocco.

Corresponding author(s). E-mail(s): anas_nour@um5.ac.ma;

Abstract

Introduction. Reuse of public metabolomics data requires that independent sampling units be identifiable; unique sample labels do not identify distinct participants. **Objectives.** To assess whether repository metadata and linked documentation establish participant-level sampling structure under prespecified, affirmative evidence rules. **Methods.** A metadata-only census of 3,582 public MetaboLights accessions; a secondary audit of factor metadata in 104 previously selected Metabolomics Workbench studies; a randomised documentary assessment of 40 of 371 unresolved MetaboLights candidates, reading deposit metadata with linked publications; and a pre-verification of 19 human deposits removed by an automated quality-control rule. Source inventories were frozen and signed before reading, grading was deterministic, extraction was AI-assisted, and the author reviewed a prespecified subset. **Results.** Of 797 human single-matrix MS or NMR studies reaching the census independence test, none met the criteria. No randomly selected study met the strict one-sample-per-participant standard (0 of 40; one-sided 95% upper bound 25 of 371 candidates). Documentation described repeated sampling in 9 of 40 studies (8 on re-review) and cell lines in 3. In 17 of 19 quality-control-excluded deposits, other fields identified study samples rather than blanks. The author's review disagreed with the AI extraction only on assigning counts to the selected assay, never on the strict result. **Conclusion.** Under prespecified documentary criteria, participant-level sampling independence was not established in the census or the randomised assessment. The findings concern what documentation establishes, not true sample independence. Deposition standards should require explicit participant-to-sample linkage.

Keywords: metabolomics, data reuse, external validation, metadata, repositories, pseudoreplication

1 Introduction

Public metabolomics deposits support method development, benchmarking and model evaluation (Haug et al, 2020; Sud et al, 2016). These uses require an analyst to identify which samples share a participant. Repeated measurements can support validation if dependence is modelled or one prespecified observation is selected per participant. Treating rows as independent without establishing their sampling units risks pseudoreplication (Hurlbert, 1984; Lazic, 2010). Likewise, deposits that share participants cannot serve as independent validation units (Riley et al, 2016).

Transcriptomic studies have documented hidden duplicates that can inflate cross-study validation and widespread sample misannotation (Waldron et al, 2016; Toker et al, 2016). Controlled-access genomic archives address linkage directly: dbGaP requires a subject–sample mapping and permits several samples per subject (National Center for Biotechnology Information, n.d.). Metabolomics audits have found uneven compliance with reporting standards, variable metadata quality, mismatched sample identifiers and limited controlled-vocabulary coverage (Spicer et al, 2017a,b; Ferreira et al, 2017; Powell and Moseley, 2021; Jarmusch et al, 2020). We are unaware of a metabolomics audit that tests whether available documentation identifies independent sampling units under a prespecified evidence rule.

We apply an affirmative evidence standard in four analyses with distinct denominators: a metadata census of 3,582 public MetaboLights accessions; a factor-metadata audit of 104 selected human Metabolomics Workbench studies; a documentary assessment of 40 randomly selected unresolved MetaboLights candidates; and a pre-verification of 19 human MetaboLights deposits excluded by a quality-control rule. The findings measure what the available documentation establishes about sampling.

2 Methods

All four analyses were metadata- and documentation-only. No measured metabolite value, data matrix, raw or derived data file was opened at any stage. Each analysis was specified in a written protocol, implemented in code with an offline test suite, and bound by SHA-256 hashes in a lock file signed by the author before execution. The locks were not registered with an external registry. Protocols, code, locks and outputs are public (Declarations).

2.1 Target property and evidence standard

The unit of interest is a deposited assay: the biological samples measured on one platform for one matrix. We ask whether the deposited metadata or linked documentation identify its independent sampling units: whether the samples come from human participants, how many distinct participants contributed, and whether any contributed more than one sample (time points, pre/post samples, several specimens, replicates entered as biological samples) or, failing that, whether a sample-to-participant mapping lets an analyst model the dependence. Evidence had to be affirmative: unique sample or source labels were bookkeeping, not evidence; counts had to be stated integers or sums over groups the source called disjoint; an enrolment figure counted only

if the text tied it to the analysed samples. Independence established in this sense is "documented"; a study whose documentation is silent may still consist of independent participants.

2.2 MetaboLights census

A locked enumeration audit fixed the universe at 3,582 accessions (REST list ∩ FTP mirror, minus two development exemplars) on 25 October 2026. The census read only investigation, sample and assay files; data files were refused by an allow-list guard. Studies were screened in a fixed order (human organism; removal of quality-control, blank, pooled and reference rows; sample-name integrity; a single matrix; the independence test). The independence test had two affirmative routes: a person-identifier column with exactly one sample per identifier (R1), or a cross-sectional design annotation plus a study-description sentence matching one of four frozen one-sample-per-participant templates, guarded against negation (R2); both required at least 50 sources. The strict criterion (L1) counted R1 and explicit R2 statements; a sensitivity count added count-only statements. Studies that failed the test but had at least 50 distinct source labels with one sample each, or at least 50 samples with a degenerate source column, became unresolved candidates for documentary assessment (Supplementary Methods S1).

2.3 Secondary audit of Metabolomics Workbench subject fields

This analysis is secondary. It reuses 104 human Metabolomics Workbench studies retained on 17 October 2026 by an earlier value-blind screen built for a different, subsequently closed research question; its criteria and exclusion counts, copied from its hashed closure record, are in Supplementary Table S1. The 104 are not a representative sample of the Workbench, and no inference is drawn about the repository as a whole. For each study we read the `factors` table returned by the Workbench REST interface (from the screen's cache; no network request) and recorded whether any factor key named a subject, patient, participant, individual, donor, person, volunteer or proband. Workbench mwTab files, which may hold an optional subject column, embed measured values and were not opened, so nothing is claimed about subject fields elsewhere in the Workbench.

2.4 Documentary assessment of a random tranche (T1)

Population and selection.

The census produced 371 unresolved MetaboLights candidates with at least 50 unique source labels (312) or at least 50 samples with a degenerate source column (59). A tranche of 40 was selected as follows. Each candidate was ranked by SHA-256 of a salt concatenated with its accession, and the first 40 were taken. The salt was the output value of a NIST Randomness Beacon (version 2) pulse, at a time fixed in the signed lock before the pulse existed (pulse 2-2005958, 30 October 2026, 15:00 UTC). No candidate was replaced. The tranche is therefore a simple random sample of 40 of the 371 candidates, and finite-population inference applies.

Sources and document freeze.

Extractors used deposited investigation files and restricted reading views of linked or accession-matched publications. The views excluded results-type material. No author was contacted. The author signed an inventory of document and view hashes before extraction, after which no source could be added (Supplementary Methods S3).

Questions and grades.

For the selected assay, extractors recorded human origin (Q1), assay membership (Q2), participant count and scope (Q3), one-sample or repeated sampling (Q4), overlap flags (Q6), and any published sample-to-participant mapping (Q7). Each answer other than *not stated* required a located verbatim quotation. Deterministic grading followed Supplementary Table S2. STRICT requires one sample per participant for a downstream analysis; Q4 and Q7 separately address sampling structure and reconstructible linkage.

Extraction, grading and adjudication.

Extraction was performed by AI sub-agents working under a written brief fixed and hashed before any record was produced (Section 2.8). The extractors were not told the deposits' biological row counts; the correspondence between a stated count and the deposited rows (rules 6–7) was computed only by the grading function. Every quotation was checked automatically for verbatim presence in the cited view. The author adjudicated every CONTRADICTED and STRICT record against the filed evidence; a rejected CONTRADICTED record became UNRESOLVED. No record was edited after grading.

2.5 Pre-verification of deposits removed by quality-control filtering

A diagnostic after the census identified 19 MetaboLights deposits with human organism values and at least 50 sample rows in which every row had been removed by the M2 quality-control rule. Because the census selected no assay for them, the four checks were applied to the biological samples listed in each deposit's sample file: V1, human participants; V2, the rows are identified as participants' samples by fields other than the sample-type label, or by a publication; V3, an integer number of biological samples stated in text (counting rows was not accepted); and V4, the sampling structure, failing on explicit repeated sampling. A study was verified if all four checks passed and failed if any check failed. Sources were the investigation, sample and assay files and named publications. The evidence set was frozen and signed before reading, extraction was blind to row counts, and quotations of sample names or individual-level rows were prohibited.

2.6 Statistical analysis

For the random tranche, the number of STRICT studies among $n = 40$ drawn without replacement from $N = 371$ bounds the number K of documentation-recoverable candidates: the one-sided 95% upper bound is the largest K with $P(X \leq x \mid N, K, n) > 0.05$

under the hypergeometric distribution (Clopper–Pearson bounds scaled by N are given for comparison). These bound candidates whose documentation meets the standard, not truly independent cohorts. All other results are counts with their denominators.

2.7 Author review of the AI-assisted extraction

The review protocol was hashed on 31 October 2026, before selecting studies. The intended independent reviewer could not be recruited, so the author performed both components on 1–2 November 2026. In component A the author re-extracted Q3 and Q4 for 10 of the 17 studies with publication text that the author had not adjudicated, selected by prespecified hash ranking. The author had not seen the original answers or quotations but knew the classifications and had seen a second model's answers for the same studies. Agreement is reported by field and is not extrapolated. In component B the original adjudicator checked all 12 CONTRADICTED records for explicit supporting statements. The review did not change the frozen grades. Supplementary Methods S5 records the deviations, including the lack of per-study completion times.

2.8 Use of AI tools

Evidence extraction for the tranche and the pre-verification was performed by sub-agents of Claude (Anthropic), working under fixed, hashed written briefs. Each sub-agent read only the sealed files of its assigned studies, produced structured records with verbatim quotations, and had to pass an automated validator that checked the schema, the verbatim presence of every quotation and the form of every count. Grades were assigned by deterministic code, not by the sub-agents. The author adjudicated every CONTRADICTED record and carried out the human validation in Section 2.7. The briefs, validators and raw sub-agent outputs are public. Claude also assisted with code and drafting under the author's direction; the author is responsible for the content.

3 Results

3.1 MetaboLights census

Of the 3,582 accessions, 3,569 reached a terminal status; 11 failed in transport and 2 returned persistent HTTP 403 responses (Table 1). Most exclusions were non-human studies (2,321). Of the human studies, 797 with a single sample matrix and an MS (714) or NMR (83) assay reached the independence test. **No study met the prespecified affirmative criteria under either the strict or the sensitivity count (L1 = 0).**

Neither route found qualifying evidence. Among the inspected records (the 658 studies whose source-node columns were tested against the R1 pattern and the 220 studies rescreened after all their rows had been classified as quality control), no qualifying person-identifier column was found. The R2 design layer passed in 18 of the 797 studies, and none of the 18 contained a qualifying one-sample-per-participant sentence. The census found 312 human studies with at least 50 distinct source labels

and one sample per source, and 59 with at least 50 samples and a degenerate source column; these 371 formed the population for the documentary assessment.

In 220 studies every sample-sheet row was classified as quality control. In 183 of them the trigger was the value `blank` in a sample-type characteristic on every row. The census read the label as written; whether the rows are in fact biological samples was examined for the human subset in Section 3.5.

Table 1 Outcome of the MetaboLights census (3,582 accessions)

Status	Rule or reason	n
Excluded	non-human (M1)	2,321
Excluded	more than one matrix (M4)	67
Excluded	>1 sample per source or identifier (M5)	60
Excluded	no MS or NMR assay	1
Unresolved	source-unique, failed M5 (312 with ≥ 50 sources)	616
Unresolved	degenerate source column (59 with ≥ 50 samples)	121
Unresolved	M2: all rows QC / ambiguous QC label / unidentified rows	220 / 66 / 7
Unresolved	duplicate sample names / organism / organism part	13 / 3 / 1
Refused	format / structure / index	61 / 10 / 2
Not terminal	transport failure / HTTP 403	11 / 2
Eligible	strict and sensitivity L1	**0**

3.2 Secondary audit of Metabolomics Workbench subject fields

None of the 104 previously selected human studies had a factor key naming a subject, patient, participant, individual, donor, person, volunteer or proband in the `factors` table. The most frequent keys were design factors: sample source (21), treatment (16), group (13), sex or gender (13), diagnosis (8). Two limitations bear on this result. The inherited repeated-measures detector did not recognise underscore or camel-case time keys (for example `POST_SURGERY_TIME`, `collectionWeek`), so dependence in these studies is undercounted. Factor keys such as cell line, cell culture medium and post-mortem interval show that a "human" species label did not by itself establish human participants. These findings concern the `factors` endpoint of 104 selected studies only.

3.3 Documentary assessment of the random tranche

Of the 40 studies, 29 had at least one publication reading view in the frozen evidence set and 11 had none. Under the prespecified procedure, no qualifying statement that each participant contributed one sample to the analysed set was identified in any reading view, and no view contained a sample-to-participant table; Q7 was "none" for all 40 studies. **No study reached the STRICT grade ($x = 0$).** Because STRICT requires a stated one-sample-per-participant structure, this count does not measure how many studies' clustering could be reconstructed from other documentation; that question is addressed by Q7 and the Q4 categories reported below. The one-sided 95% hypergeometric upper bound on the number of the 371 candidates whose documentation meets the standard is 25 (Clopper–Pearson: 26.8).

Table 2 Final grades of the 40 tranche studies after adjudication

Grade (reason)	n
STRICT	0
COUNT (sensitivity only)	5
CONTRADICTED: repeated sampling in the chosen assay	9
CONTRADICTED: samples not from human participants (cell lines)	3
UNRESOLVED: adjudication rejected	3
UNRESOLVED: assay membership not established	6
UNRESOLVED: no assay-scoped participant count	5
UNRESOLVED: stated count below / above deposited rows	3 / 3
UNRESOLVED: count supported only by the deposit	3

Where documentation addressed sampling, it often described dependence. Nine studies (22.5%) documented repeated sampling of the same participants within the chosen assay (serial time points, pre/post-treatment samples, follow-up visits, several cysts per patient, up to 56 time points per participant), and three (7.5%) consisted of cultured cell lines rather than participant samples. Together these account for the 12 CONTRADICTED classifications after adjudication. All twelve had entered the candidate pool through unique or degenerate source labels. Under the author's re-review (Section 3.4) one of the nine repeated-sampling classifications is not explicitly supported, giving 8 of 40 (20%) as a sensitivity figure.

Stated counts frequently failed to reconcile with the deposits. Among the 13 studies not finally graded CONTRADICTED that reported an assay-scoped participant count with a publication source, 7 matched the number of deposited biological rows and 6 did not (3 lower, 3 higher). None of the 7 had one-sample-per-participant evidence. The candidate pool also contained four separately deposited ionisation-mode assays of one sample set (MTBLS1122, MTBLS1124, MTBLS1129, MTBLS1130), so the number of candidates overstates the number of distinct studies.

The result does not depend on unread documents or on adjudication. Three named publications could not be obtained; if all three had supplied STRICT evidence, x would be 3 and the upper bound 65. Adjudication can only confirm or reject CONTRADICTED and STRICT grades, and a rejection yields UNRESOLVED.

3.4 Author review of the AI-assisted extraction

Component A.

Q3 scope agreed in 6 of 10 studies, Q3 count in 4 of 4 where both chose the assay-scoped category, and Q4 in 10 of 10 (both "not stated" throughout, a shared null that does not test sensitivity for repeated sampling). All four Q3 disagreements (MTBLS718, 11406, 2774, 13787) ran one way: the author declined to assign a stated count to the selected assay where the extractor had, because the documents did not say which sample set the deposit held or gave a sample rather than a participant count. The author found nothing the extractor had missed. Three of the four records were already UNRESOLVED; the fourth (MTBLS11406) would move from COUNT to UNRESOLVED, reducing the sensitivity-only STRICT + COUNT figure from 5 to 4. No disagreement bears on the STRICT count.

Component B.

Of the 12 CONTRADICTED records, 11 were confirmed by an explicit statement in the cited source (the sentence the extractor had cited), none was disputed, and one (MTBLS10972, deposit text only) was judged unresolved: the deposit describes a post-meal time course without stating that multiple specimens from the same individuals entered the assay. As a sensitivity figure, documented repeated sampling is 8 of 40 (20%) rather than 9, and the CONTRADICTED total 11 rather than 12; the frozen figures in Table 2 remain primary.

3.5 Deposits removed by quality-control filtering

In 18 of the 19 human deposits whose rows had all been removed by the quality-control rule, every sample row carried `Characteristics[Sample type] = experimental blank`; in the remaining deposit, 94 of 95 rows did. In 17 of the 19, other deposited fields identified the rows as the study's biological samples rather than blanks (V2 passed): disease-group or treatment factor values on the same rows, sample-collection protocols, or the study description. In 16 of these 17 the same fields describe participant specimens; in MTBLS15165 they describe stored platelet units whose donors are not mentioned, so the human origin of the material is not established (V1 not established). Supplementary Table S3 lists the supporting fields per deposit. In these 17 the sample-type annotation conflicts with the other fields. Three deposits were verified (all four checks passed, with stated sample counts of 368, 144 and 85), eight failed (cell lines or non-human rows in two, pooled material in two, repeated sampling in four) and eight were unverifiable, mostly because no sample count was stated.

Two further inconsistencies were found. In both assay files of MTBLS14421, the extract, assay and normalisation names of 50 of 53 rows carry the label of a different sample, and in 30 rows that label belongs to the other study group (15 control rows carry patient labels and 15 patient rows carry control labels). This is a linkage inconsistency between the sample sheet and the assay table; the deposit does not show which label is correct, and we make no claim about the published analysis. MTBLS14358 and MTBLS14376 share a title and a cohort description; whether their sample sets overlap is not established from the deposits.

3.6 Implementation limitations

The frozen implementation had three limitations, each disclosed in the result records and described in the Supplementary Note: one reading view included a differential-metabolite table whose caption escaped the exclusion pattern (nothing was cited from it); the PDF view heuristic did not recognise one heading variant (no graded record depended on it); and the count check accepted a deposit integer when the participant-count question also cited a publication (one record, whose publication stated the component group sizes).

4 Discussion

The four analyses answer different documentary questions. The census and random assessment tested one-sample-per-participant criteria, the Workbench audit tested one factor-metadata endpoint, and the pre-verification tested sample classification. The results apply to their respective frames and do not estimate repository-wide prevalence.

Participant linkage.

The nine tranche studies with documented repeated sampling had no qualifying participant-to-sample mapping. Without it, an analyst cannot account for within-participant dependence. At the study level, counting accessions would count four ionisation-mode assays of one sample set (MTBLS1122, MTBLS1124, MTBLS1129, MTBLS1130) as four validation units. Two pre-verification deposits (MTBLS14358, MTBLS14376) shared a title and cohort description, but their documentation did not establish whether their samples overlapped.

Sample classification.

The census classified every row as quality control in 220 studies, including 183 where every sample-type value was `blank`. The 19 human pre-verification deposits came from that subset. Other fields identified study samples in 17 and participant specimens in 16. The label appears to be a template default. Omitting pre-filled values would reduce this ambiguity. The inconsistent assay-to-sample linkage in MTBLS14421 was detectable from the deposit but could not be resolved from it.

Evidence standards and limitations.

Unique labels alone would have qualified hundreds of census studies, although the random tranche included 9 studies with repeated sampling and 3 with cell lines. Screening should prespecify affirmative evidence and report failure categories. The documentary bound applies only to the 371 filtered candidates. Publication text was unavailable for 11 of 40 sampled studies. STRICT requires one sample per participant, so zero STRICT does not imply zero reconstructible studies. The Workbench result applies only to the audited endpoint in the selected studies. The author review supports the zero-STRICT finding but is not independent human validation. Component A followed exposure to a second model's answers, and shared negative Q4 assessments do not measure sensitivity for repeated sampling. Component B was a non-blind check by the original adjudicator. The frozen implementation limitations are reported in Section 3.6. Missing documentation does not establish true sample dependence.

Recommendations.

Repositories should require a participant identifier or subject-to-sample mapping for each biological sample, flag deposits that share sample sets, and avoid pre-filled sample-type values. STROBE-MetEpi, currently under review, asks authors to report sampling, collection, validation populations, counts, timing and repository access, but does not require a machine-readable mapping from deposited samples

to participants (Kelly et al, 2026). A deposit-level mapping would complement those manuscript recommendations and support reuse of repeated measurements. Methodological studies such as this audit fall outside the guideline's scope.

Declarations

- Funding: This research received no specific funding from any funding agency in the public, commercial or not-for-profit sectors.
- Competing interests: The author declares no competing interests.
- Ethics: the study analysed publicly deposited metadata and published documents only; no participants were recruited, no biological samples were handled, and no participant-level data were accessed or recorded. Ethical approval was not required.
- Use of AI tools: see Section 2.8.
- Supplementary Information: Supplementary Methods S1–S5 and Supplementary Tables S1–S3 (file `audit_supplement.pdf`).
- Data and code availability: protocols, code, locks, sealed evidence inventories, evidence records, grades and result records are public in the repository https://github.com/Anas-Enoch/Pathway_Identifiability (commits `b5cd507`, `3d376d0`, `195a60d`, `566af9d`; author-review records in `c27ec53`). The submission source is archived under `audit-paper-v0.8`, and the earlier full-length version under `audit-paper-v0.7`. Full texts of publications are not redistributed; their SHA-256 values and retrieval URLs are recorded.
- Author contribution: A.E. designed the study, executed the protocols, adjudicated and reviewed the records, and wrote the manuscript.

References

Ferreira JD, Inácio B, Salek RM, et al (2017) Assessing public metabolomics metadata, towards improving quality. Journal of Integrative Bioinformatics 14(4). https://doi.org/10.1515/jib-2017-0054

Haug K, et al (2020) MetaboLights: a resource evolving in response to the needs of its scientific community. Nucleic Acids Research 48(D1):D440–D444. https://doi.org/10.1093/nar/gkz1019

Hurlbert SH (1984) Pseudoreplication and the design of ecological field experiments. Ecological Monographs 54(2):187–211. https://doi.org/10.2307/1942661

Jarmusch AK, et al (2020) ReDU: a framework to find and reanalyze public mass spectrometry data. Nature Methods 17:901–904. https://doi.org/10.1038/s41592-020-0916-7

Kelly RS, Reinke SN, et al (2026) A reporting guideline for observational studies in metabolomic epidemiology: explanation and elaboration of the STROBE-MetEpi checklist. Research Square preprint, 25 June 2026, under review

Lazic SE (2010) The problem of pseudoreplication in neuroscientific studies: is it affecting your analysis? BMC Neuroscience 11:5. https://doi.org/10.1186/1471-2202-11-5

National Center for Biotechnology Information (n.d.) dbGaP submission guide. https://www.ncbi.nlm.nih.gov/gap/docs/submissionguide, accessed 31 October 2026

Powell CD, Moseley HNB (2021) The mwtab Python library for RESTful access and enhanced quality control, deposition, and curation of the Metabolomics Workbench data repository. Metabolites 11:163. https://doi.org/10.3390/metabo11030163

Riley RD, et al (2016) External validation of clinical prediction models using big datasets from e-health records or IPD meta-analysis: opportunities and challenges. BMJ 353:i3140. https://doi.org/10.1136/bmj.i3140

Spicer RA, Salek R, Steinbeck C (2017a) Compliance with minimum information guidelines in public metabolomics repositories. Scientific Data 4:170137. https://doi.org/10.1038/sdata.2017.137

Spicer RA, Salek R, Steinbeck C (2017b) A decade after the metabolomics standards initiative it's time for a revision. Scientific Data 4:170138. https://doi.org/10.1038/sdata.2017.138

Sud M, et al (2016) Metabolomics Workbench: an international repository for metabolomics data and metadata, metabolite standards, protocols, tutorials and training, and analysis tools. Nucleic Acids Research 44(D1):D463–D470. https://doi.org/10.1093/nar/gkv1042

Toker L, Feng M, Pavlidis P (2016) Whose sample is it anyway? Widespread misannotation of samples in transcriptomics studies. F1000Research 5:2103. https://doi.org/10.12688/f1000research.9471.1

Waldron L, Riester M, Ramos M, et al (2016) The Doppelgänger effect: hidden duplicates in databases of transcriptome profiles. Journal of the National Cancer Institute 108(11):djw146. https://doi.org/10.1093/jnci/djw146